

# ARYAN SHARMA

## Mobile Engineer (iOS · Android · Cross-Platform)

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### SUMMARY

Mobile engineer with real cross-platform native work: a single Rust engine bridged to iOS (Swift XCFramework) and Android (Kotlin JNI) via UniFFI, a BLE GATT protocol design that works around the quirks of both CoreBluetooth and BluetoothGattServer, and native-feel desktop/mobile products (Tauri, PyQt) with one-line distribution. 10,700+ LOC of mobile systems code, including fixes only possible on the platform side.

### TECHNICAL SKILLS

- iOS: Swift · CoreBluetooth (Peripheral + Central) · CryptoKit · Secure Enclave · XCFramework · App Intents · Tauri on macOS (Core Graphics)
- Android: Kotlin · BluetoothGattServer/GattClient · Android Keystore · java.security · JNI · XCFramework/JNI interop · Gradle
- Cross-Platform: Mozilla UniFFI (Rust→Swift/Kotlin FFI) · byte-identical cipher parity across stacks · shared protocol design · BLE (MTU negotiation, CCCD, fragmentation)
- Product: mobile-first React/Next.js · WebView-free native capture paths · one-line installers (curl) · PyInstaller/gradle packaging · App Store-adjacent licensing (HWID/UUID binding)
- Languages: Swift · Kotlin · Rust · TypeScript · Python

### EXPERIENCE

#### Cinntra Infotech — Systems Engineer · 2026 – Present

- Built CIRA, a RAG agent translating natural-language requests into audited SAP HANA queries and write operations, giving non-technical users direct access to enterprise data
- Built the point-of-sale system backend: transaction recording, merchant reconciliation, and offline synchronization

#### Niswa Engineering — Technical Consultant & Automation Engineer · 2024 – 2026

- Built asynchronous AI agents (TypeScript) that automated 80%+ of a content generation and scheduling pipeline
- Built the company website in Astro (95+ Google Lighthouse score); set up cloud infrastructure, API integrations, and event-driven automation workflows

#### Sudarshana Foundation — Full-Stack & Automation Engineer · Contract

- Designed and built the organization's first centralized web platform, digitizing paper-based workflows into software
- Built database systems handling 5,000+ monthly transactions

### PROJECTS

#### Astral — Cross-Platform BLE Mesh SDK (Rust core → iOS + Android)

- Built one Rust engine (astral-core: cryptography, MTU slicing, spray-and-wait routing) and bridged it to both mobile stacks with Mozilla UniFFI — Swift XCFramework for iOS, Kotlin via JNI for Android — so the mesh protocol is byte-identical on every device (10,700+ LOC total).
- Solved the iOS↔Android BLE interop problems platform by platform: manual CCCD descriptor (0x2902) writes on Android Centrals to correctly subscribe to iOS Peripheral ACKs (the "merchant did not respond" bug), sequential onCharacteristicWrite queues replacing racy timer-based writes, and dynamic MTU negotiation to 512 B with fragmentation for 200–500 B payloads.
- Implemented hardware-backed identity on both platforms: iOS Secure Enclave and Android Hardware Keystore P-256 keypairs, with ChaCha20-Poly1305 AEAD parity between Apple CryptoKit and java.security down to the byte.
- Shipped demo apps on both platforms (Android wallet + QR scanner screens, iOS app) plus a Next.js/Supabase reconciliation backend for when devices reconnect.
- Tech: Rust, Swift, Kotlin, UniFFI, BLE GATT, Noise Protocol — [github.com/Aryan-Protein-Vala/Astral-Offline](https://github.com/Aryan-Protein-Vala/Astral-Offline) · [github.com/Aryan-Protein-Vala/Astral-Network](https://github.com/Aryan-Protein-Vala/Astral-Network)

#### MYLO OS — Native Desktop Agent (macOS + Windows)

- Built a Tauri 2 (Rust) agent with first-class native integrations on both desktop OSes: zero-latency screen capture (macOS Core Graphics / Windows Graphics Capture), human-paced input injection, global hotkeys, and OS-level stealth (NSWindow.sharingType / WDA\_EXCLUDEFROMCAPTURE) so the overlay is invisible to screen share — ~35 MB RSS.
- Tech: Rust, Tauri 2, Swift (macOS surface), TypeScript — [github.com/Aryan-Protein-Vala/MyloOS](https://github.com/Aryan-Protein-Vala/MyloOS)

#### Snag — Native macOS App with One-Line Distribution

- Built a native-feel macOS utility in Python + PyQt6, packaged with PyInstaller: global-hotkey floating widget (frameless, matte dark mode), screenshot hub, real-time downloads tracker, 15-slot clipboard history, and drag-and-drop out into any app — with a curl one-liner installer that installs to /Applications and handles Gatekeeper safely.
- Tech: Python, PyQt6, PyInstaller, Next.js, Supabase — [github.com/Aryan-Protein-Vala/Snag](https://github.com/Aryan-Protein-Vala/Snag)

#### DermaOS — Mobile-First Consumer Product (live)

- Built the mobile-first product experience for an AI skincare app: scan-to-report flow, onboarding, modals, and marketplace UI tuned for small screens and Indian market context (INR pricing, UPI via Razorpay); live at [dermaos.vercel.app](https://dermaos.vercel.app).
- Tech: Next.js 16, Tailwind, shadcn/ui — [github.com/Aryan-Protein-Vala/DermaOS](https://github.com/Aryan-Protein-Vala/DermaOS)

### CERTIFICATIONS

- Frontend Development Certification — Sheryians Coding School
- Data Science & AI/ML Bootcamp Certification — Udemy